Scalability & Future Plan

Date	21 October 2026
Team ID	TDS-AI-2026
Project Name	Personalized Travel Itinerary Planner
Maximum Marks	4 Marks

Project Title

Personalized Travel Itinerary Planner

Current Scalability Assessment

Backend Scalability

Aspect	Current State	Scalability Rating
Stateless API design	FastAPI is fully stateless — no in-memory session state	Excellent
Database connection	Per-request Supabase client (no connection pooling overhead)	Good
LLM calls	Synchronous but isolated per request — no blocking global state	Good
Horizontal scaling	Multiple Render instances can run behind a load balancer	Excellent
Caching	No caching layer currently (Redis could be added)	Needs improvement

Database Scalability

Aspect	Current State	Scalability Rating
PostgreSQL via Supabase	Managed PostgreSQL with automatic backups	Excellent
Row-Level Security	Enforced at DB level — no additional app-layer code needed at scale	Excellent
Indexes	Optimized indexes on owner_id, trip_id, day_id, position	Good
Connection pooling	Supabase provides PgBouncer connection pooling	Good

Frontend Scalability

Aspect	Current State	Scalability Rating
Vercel CDN	Global CDN with edge caching	Excellent

Aspect	Current State	Scalability Rating
React Query caching	Client-side caching reduces redundant API calls	Excellent
Bundle size	Vite build with tree-shaking	Good

Short-Term Improvements (1-3 months)

1. Async AI Generation

- **Current**: AI generation is synchronous (10-30 seconds blocks the response)
- **Improvement**: Use FastAPI background tasks or Celery with Redis for async processing
- **Benefit**: Instant response to user with status polling, better UX for large trips

2. Caching Layer

- Add Redis caching for frequently accessed trip data
- Cache AI responses for similar trip configurations to reduce API costs
- Implement cache invalidation on trip/activity updates

3. Rate Limiting

- Add per-user rate limiting for AI generation endpoints (e.g., 10 trips/day)
- Prevent API abuse and control LLM costs
- Use Redis for distributed rate limit counters

4. Collaborative Planning (Already Partially Implemented)

- The database already has a `trip_members` table with editor/viewer roles
- Implement the API endpoints and UI for inviting collaborators
- Add real-time updates when collaborators make changes (Supabase Realtime)

Medium-Term Improvements (3-6 months)

5. Map Integration

- Add Google Maps or Mapbox to show activities as map pins
- Show route planning between activities within a day
- Allow users to click map pins to view activity details

6. Export Features

- Export itinerary to PDF directly from the UI
- Export to Google Calendar / iCal format
- Share itinerary via public URL

7. Multi-Model Support

- Support GPT-4o and Claude as additional AI providers
- Allow users to select their preferred AI model
- A/B test different models for quality comparison

8. Activity Suggestions

- AI-powered "add more activities" feature that suggests activities based on remaining budget and time

- Integration with travel APIs (Google Places, TripAdvisor) for verified location data

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Long-Term Vision (6+ months)

9. Mobile App

- React Native or Flutter mobile application
- Offline itinerary access
- GPS-based activity check-ins

10. Social Features

- Share and discover public itineraries from other travelers
- Community ratings and reviews for generated itineraries
- Clone popular itineraries and customize them

11. Booking Integration

- Direct booking links for hotels, restaurants, and attractions
- Price tracking and alerts for activities
- Integration with booking platforms (Booking.com, Airbnb, GetYourGuide)

12. Enterprise Features

- Team/organization accounts for travel agencies
- Custom branding options
- Analytics dashboard for itinerary performance

Infrastructure Evolution

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Current:                        Future:
Vercel (Frontend)               Vercel + CDN Optimization
     |                               |
Render (FastAPI)    ------>     Render (API) + Redis Cache + Background Workers
     |                               |
Supabase (PostgreSQL)           Supabase (Primary) + Read Replicas
     |                               |
Gemini/Groq (LLM)               Multi-LLM Router with Cost Optimization
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